Framing:

Translates the physical layer's raw bit stream into discrete units called frames

- How can the receiver detect frame boundaries?
- That is, how can the receiver recognize the start and end of a frame?

Four methods:

1. Length Count
2. Bit Stuffing
3. Character stuffing
4. Pulse encoding

Character count

Field in header gives no. of chars in frame.
Shown in (a) below. Char count includes the counting character itself:

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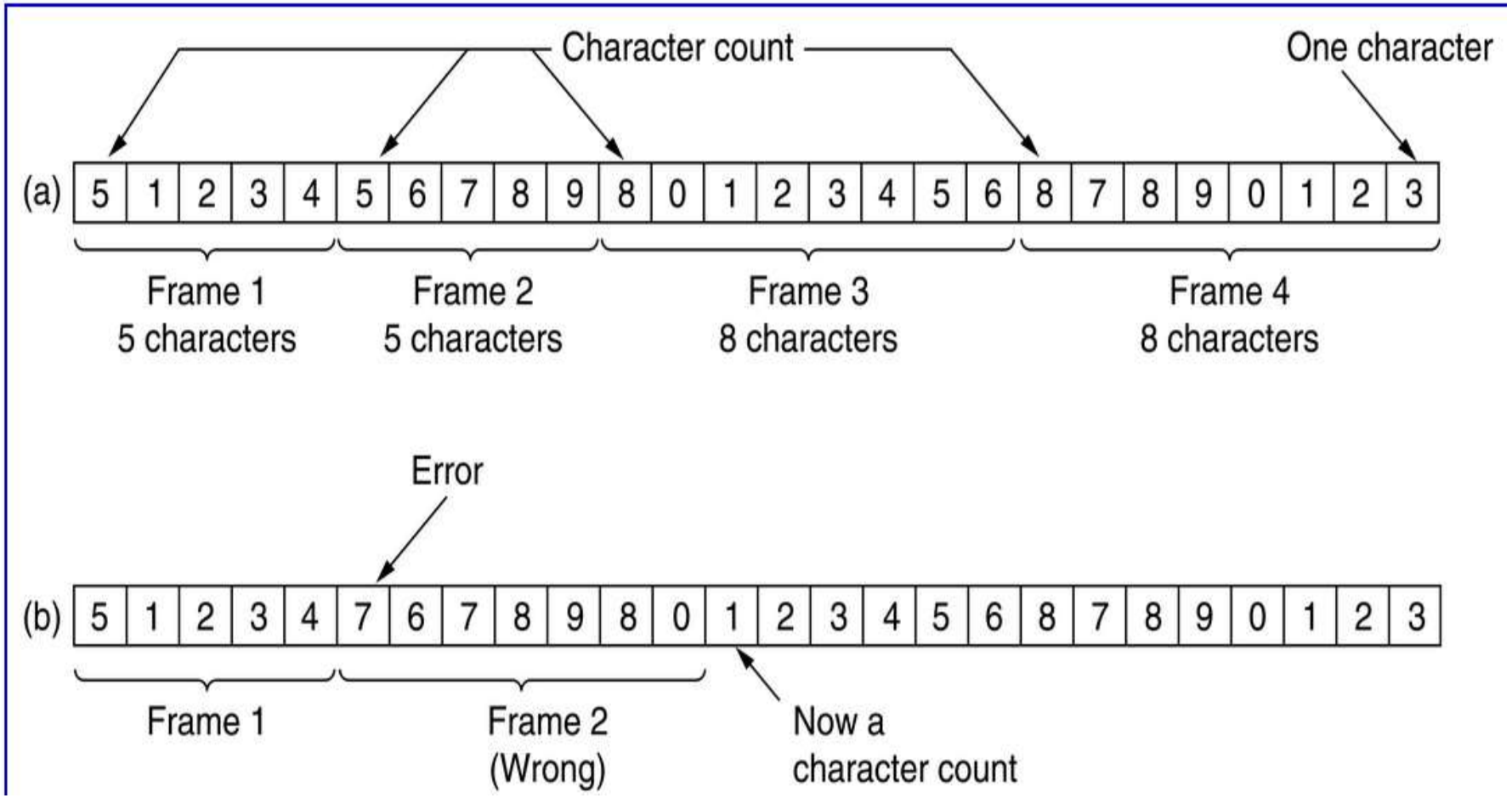
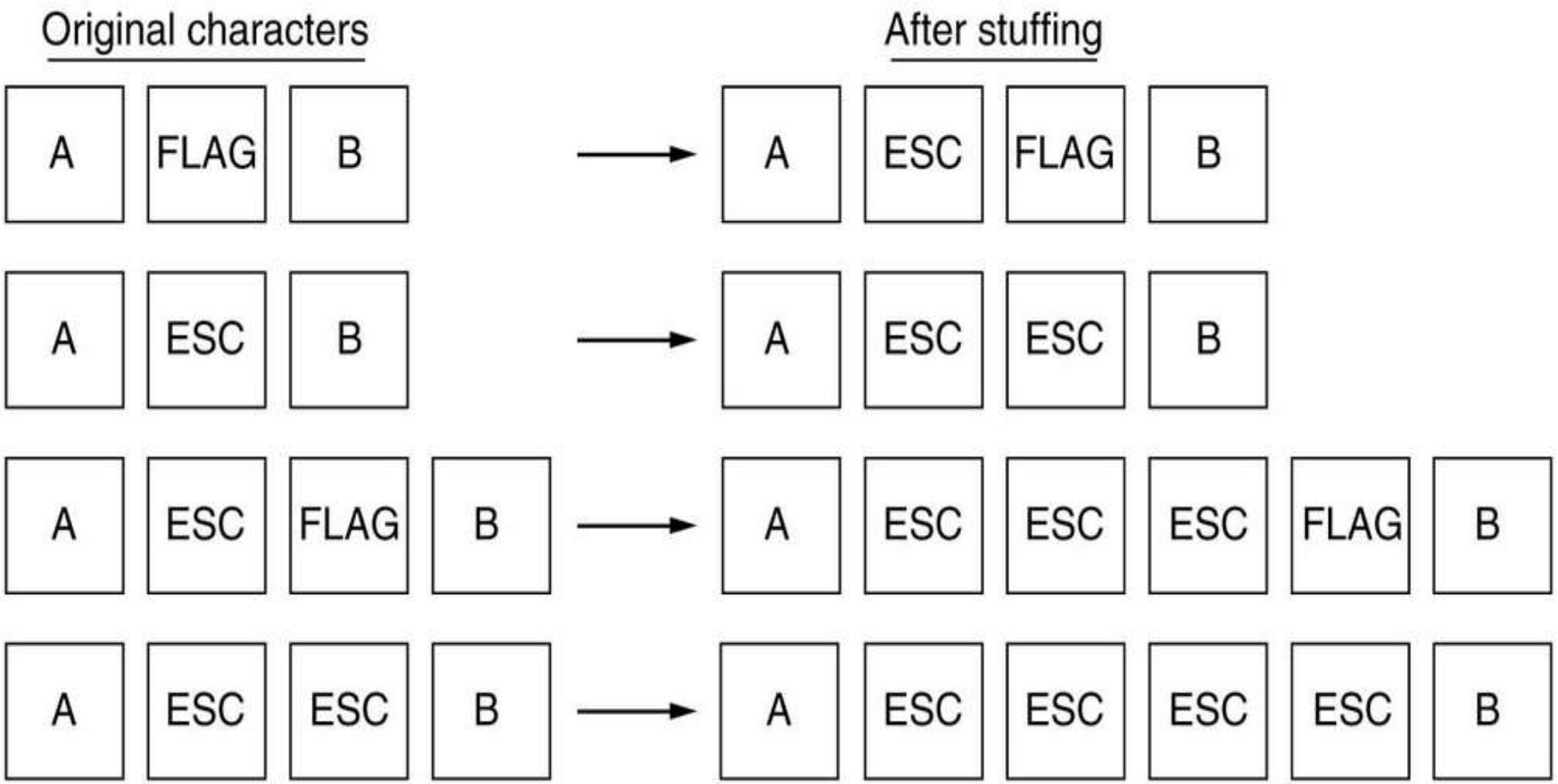


Start and end bytes (with byte stuffing)

Each frame starts with special start and end bytes (**flag bytes**). Here will imagine it as same byte, FLAG.
After error, can always find start of next frame.
See (a) below:


Original characters
After stuffing
A FLAG B
A ESC FLAG B
A ESC B
A ESC ESC B
A ESC FLAG B
A ESC ESC ESC FLAG B
A ESC ESC B
A ESC ESC ESC ESC B

Start and end flags (with bit stuffing)

Byte stuffing specifies char format (e.g. 8 bits per char).
To allow arbitrary no. of bits per char, use stuffing at *bit*-level rather than at byte-level.

Each frame begins and ends with bit pattern 01111110 (6 1's)
If 5 1's in a row in data, stuff a 0 in so will never be 6 in a row.
Stuff it in always - *whether the next char was going to be a 1 or not.*
De-stuffer removes the 0's after any 5 1's.

(a) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

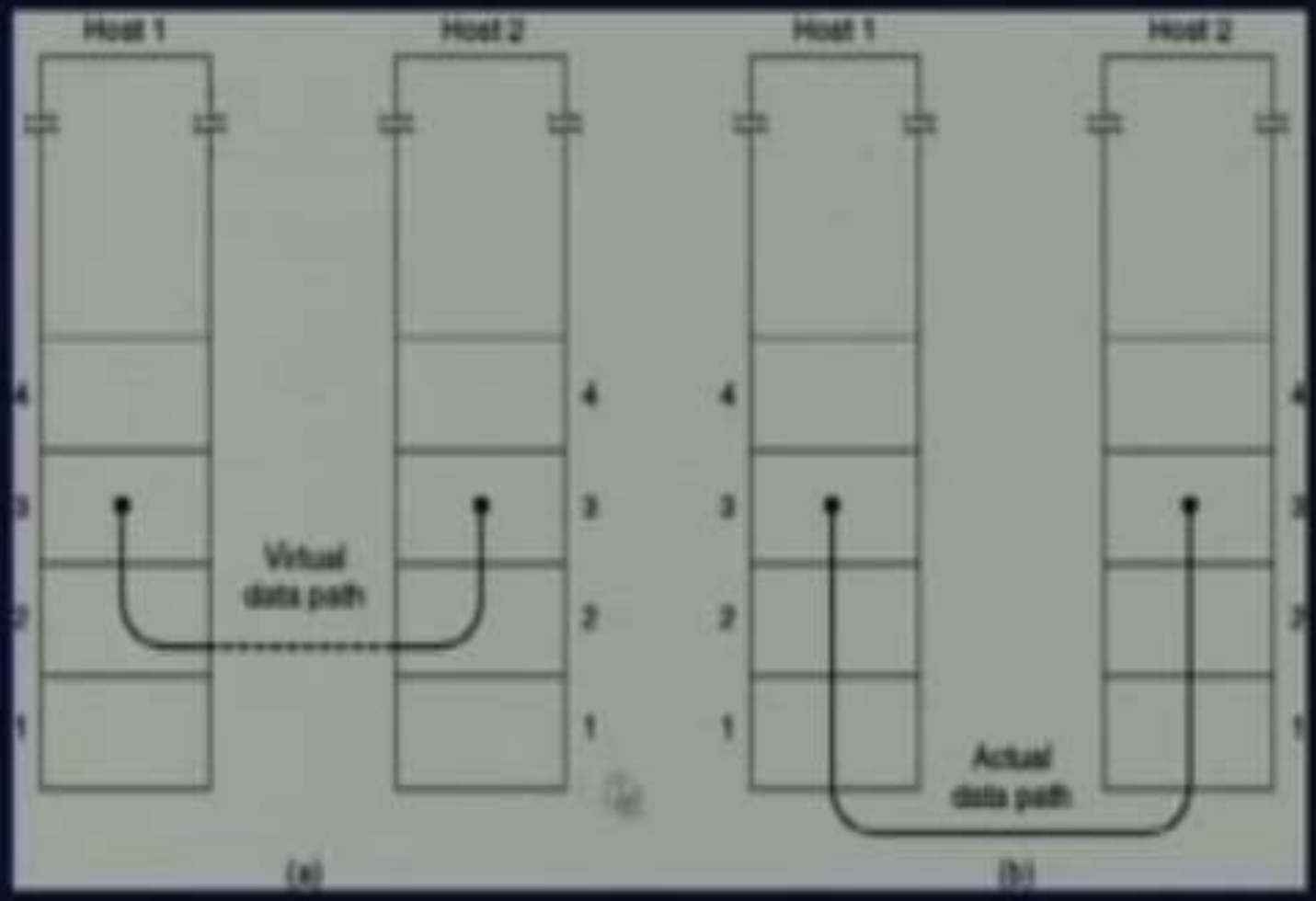
(b) 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 1 0 0 1 0

Stuffed bits

(c) 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0

- The main tasks of the data link layer are:
  - Transfer data from the network layer of one machine to the network layer of another machine
  - Convert the raw bit stream of the physical layer into groups of bits ("frames")

Virtual & Actual Communication.



Why Data Link Control?

- Frame synchronization: The beginning and end of a data block (called *frame*) should be recognizable.
- Flow control: The sender should not send frames at a rate faster than the receiver can receive and process them.
- Error control: Any bit errors introduced by the transmission system should be corrected.

Why Data Link Control?

- Addressing: On a multipoint link (line LAN), the identity of sender and receiver must be specified.
- Control and data on same link: The receiver must be able to distinguish control information from the data being transmitted.
- Link management: Procedures for the management of the initiation, maintenance, and termination of a sustained data exchange

Link Layer Services

- Framing, link access:
  - encapsulate datagram into frame, adding header, trailer
  - channel access if shared medium
  - "MAC" addresses used in frame headers to identify source, dest
    - different from IP address!

Link Layer Services

- Flow Control:
  - pacing between adjacent sending and receiving nodes
- Error Detection:
  - errors caused by signal attenuation, noise.
  - receiver detects presence of errors:
    - signals sender for retransmission or drops frame

DLL - Type of service

Acknowledged connection-oriented

- Ensures that
  - All frames are received
  - Each is received exactly once

The service is accomplished using SIMPLEX, HALF DUPLEX OR FULL DUPLEX channels.

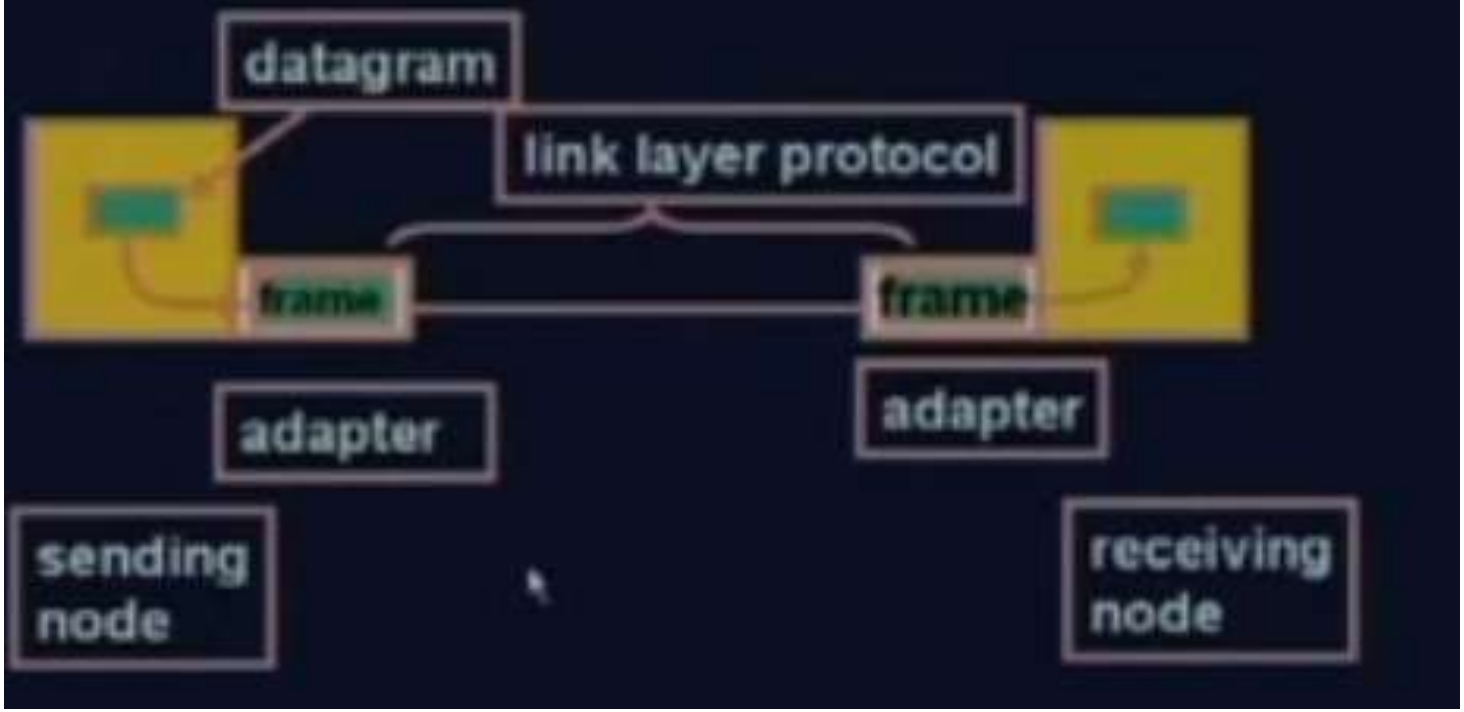
DLL - Type of service

- Unacknowledged connectionless
  - No attempt to recover lost frame
  - Suited for low error rate networks or for fault tolerant applications such as voice
- Acknowledged connectionless
  - Each frame is acknowledged by the receiver
  - Suited for unreliable channels

Types of Services

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Adaptors Communicating



Adaptors Communicating

- link layer implemented in "adaptor" (aka NIC)
  - Ethernet card, PCMCI card, 802.11 card
- sending side:
  - encapsulates datagram in a frame
  - adds error checking bits, rdt, flow control, etc.

Adaptors Communicating

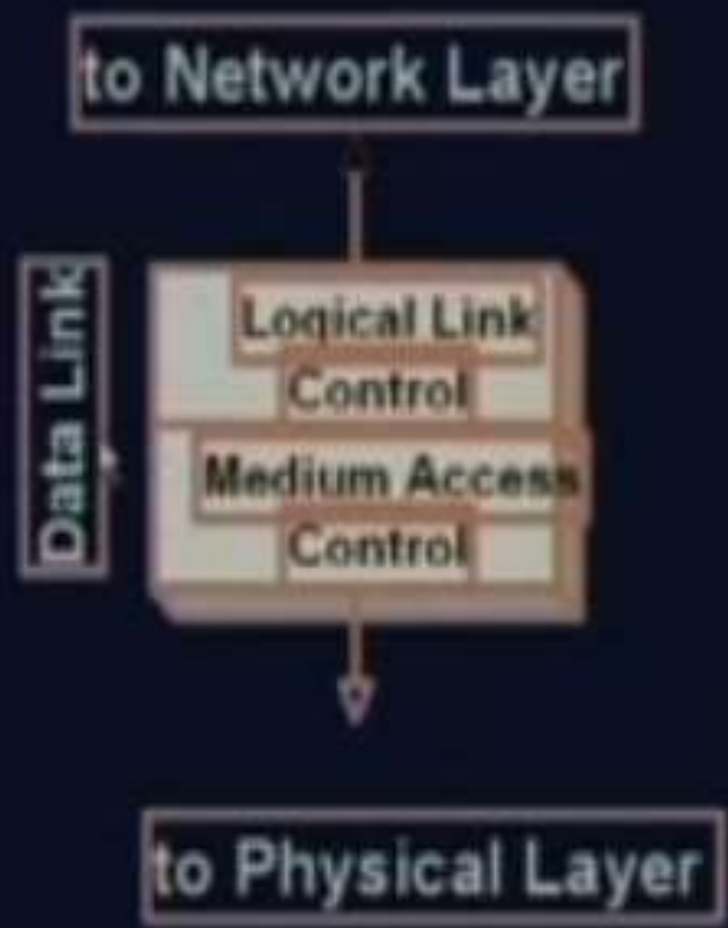
- receiving side
  - looks for errors, rdt, flow control, etc
  - extracts datagram, passes to receiving node
- adapter is semi-autonomous and communicates directly with link & physical layers

MAC and LLC

- In any broadcast network, the stations must ensure that only one station transmits at a time on the shared communication channel
- The protocol that determines who can transmit on a broadcast channel are called Medium Access Control (MAC) protocol

MAC and LLC

- The MAC protocol are implemented in the MAC sublayer which is the lower sublayer of the data link layer
- The higher portion of the data link layer is often called Logical Link Control (LLC)



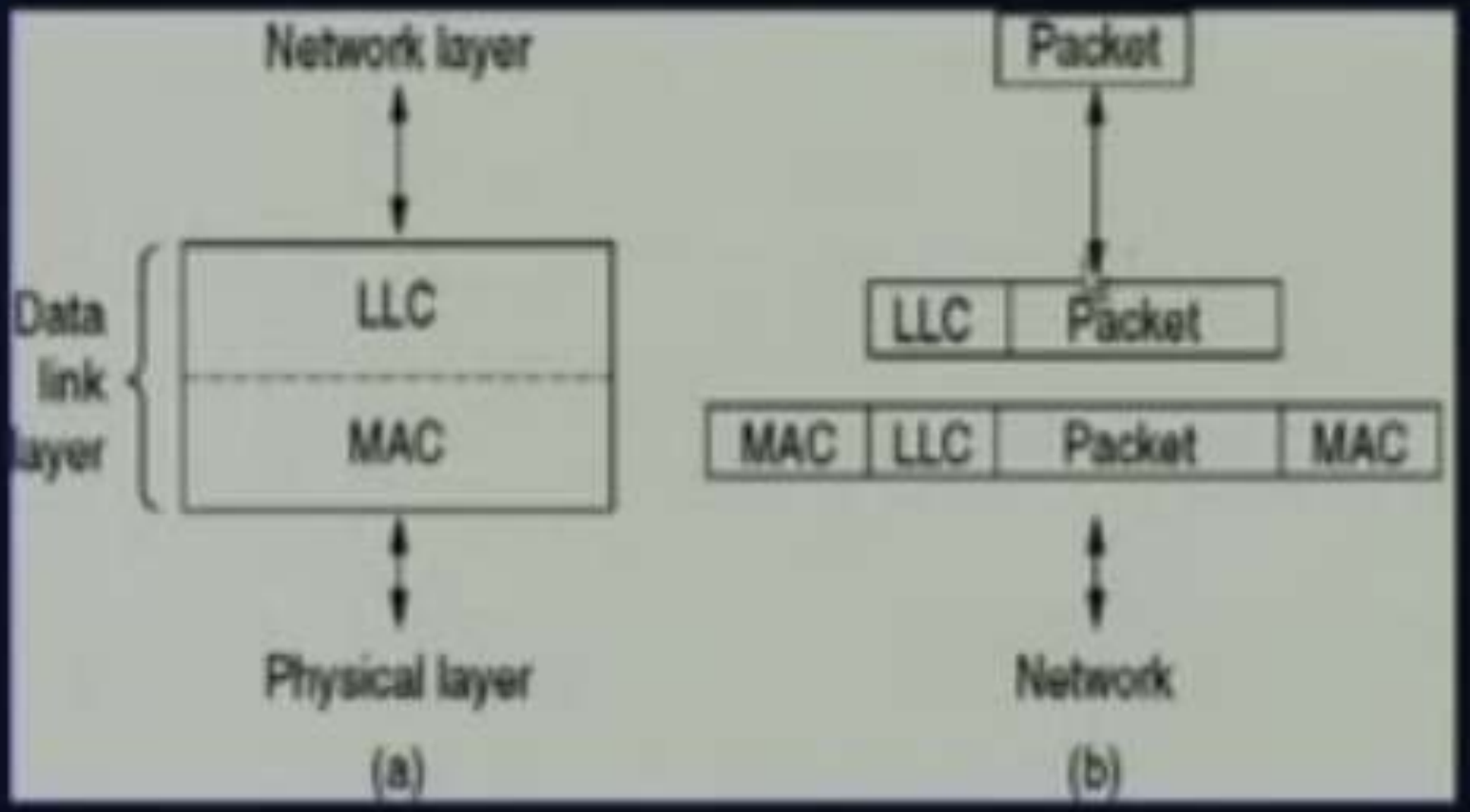
IEEE 802 Standards

- IEEE 802 is a family of standards for LANs, which defines an LLC and several MAC sublayers

Overview of IEEE LAN Standards

• 802.1	Overview
• 802.2	Logical Link Control
• 802.3 (Ethernet)	CSMA/CD
• 802.4	Token Bus
• 802.5	Token Ring
• 802.6	Distributed Queue Dual Bus
• FDDI	Fiber Distributed Data Interface

IEEE 802.2: Logical Link Control



(a) Position of LLC. (b) Protocol formats.

Logical Link Control(LLC)

- The 802 LANS offer a best effort dataframe services
- Error control and flow control are handled by LLC
- LLC runs on all three 802 LANs and hides the differences to the network layer.

Logical Link Control(LLC)

- LLC adds its header to the network layer packet. It contains sequence and acknowledgement numbers
- Resulting structure goes into the payload of 802.x frame for transmission

LLC Types and Classes

- Type 1 operation supports unacknowledged connectionless service
- Type 2operation supports connection-mode service
- Type 3 operation supports acknowledged connectionless service

LLC Protocol Data Units

- Information
  - used to carry user data. The control field includes a 7-bit sequence number , N(S), associated with this PDU; it also include a piggybacking acknowledgement sequence number, N(R)
- Unnumbered
  - Various protocol control PDUs. The 5-bit M field indicates which PDU

LLC Protocol Data Units

- Supervisory
  - used for flow and error control. It includes an acknowledgement sequence number and a 2-bit S field to distinquish three different PDUs: receive ready (RR), receive not ready (RNR), and reject (REJ)

Type 1 Operation

- Type 1 operation supports the unacknowledged connectionless service.
- The UI PDU is used to transfer user data. There is not ack, flow control, or error control.
- The XID and TEST PDUs support management functions associated with all three types of operation.

Type 2 Operation

- Type 2 operation supports the connection-mode service.
- Type 2 operation involves three phases:
- connection establishment,
- data transfer, and
- connection termination.

Type 3 Operation

- With type 3 operation, each PDU transmitted is acknowledged.
- A new unnumbered PDU, the Acknowledged Connectiuonless (AC) Information PDU is defined.
- User data are send in AC command PDUs and must be ack'ed using an AC response PDU.

Type 3 Operation

- To guard against lost PDUs, a 1-bit sequence number is used.
- The sender alternates the use of 0 and 1 in its AC command pDU,
- and the receiver responds with an AC PDU with the opposite number of the corresponding command.
- Only one PDU in each direction may be outstanding at any time.

Frame Synchronization

- When data are transferred from the transmitter to the receiver, unless steps are taken to provide synchronization, the receiver may start interpreting the data erroneously.
- Two common approaches:
  - Asynchronous Transmission
  - Synchronous Transmission

Asynchronous Transmission

- Data are transmitted one character at a time (5-8 bits).
  - Timing or synchronization must only be maintained within each character.
  - The receiver has the opportunity to resynchronize at the beginning of each new character.

Asynchronous Transmission

When no character is being transmitted, the line between transmitter and receiver is in an idle state.

– Start, stop, and parity bits.

Synchronous Transmission

- Here, a block of bits is transmitted in a steady stream without start and stop codes.
  - The block may be arbitrarily long.
  - To prevent timing drift between transmitter and receiver, clock signal is embedded in the data signal (e.g. Manchester encoding).

Synchronous Transmission

- There is another level of synchronization required, so as to allow the receiver to determine the beginning and end of a block of data.
  - Every block begins with a preamble bit pattern, and generally ends with a postamble bit pattern

Synchronous Transmission

- A typical synchronous frame format:
  - 8-bit flag (preamble)
  - Control fields
  - Data field
  - Control fields
  - 8-bit flag (postamble)

Synchronous Transmission

- For sizable blocks of data, synchronous transmission is far more efficient than asynchronous mode.
  - Asynchronous transmission requires 20% or more overhead.
  - The control information, preamble and postamble in synchronous transmission are typically less than 100 bits.

Framing:

Translates the physical layer's raw bit stream into discrete units called frames

- How can the receiver detect frame boundaries?
- That is, how can the receiver recognize the start and end of a frame?

Four methods:

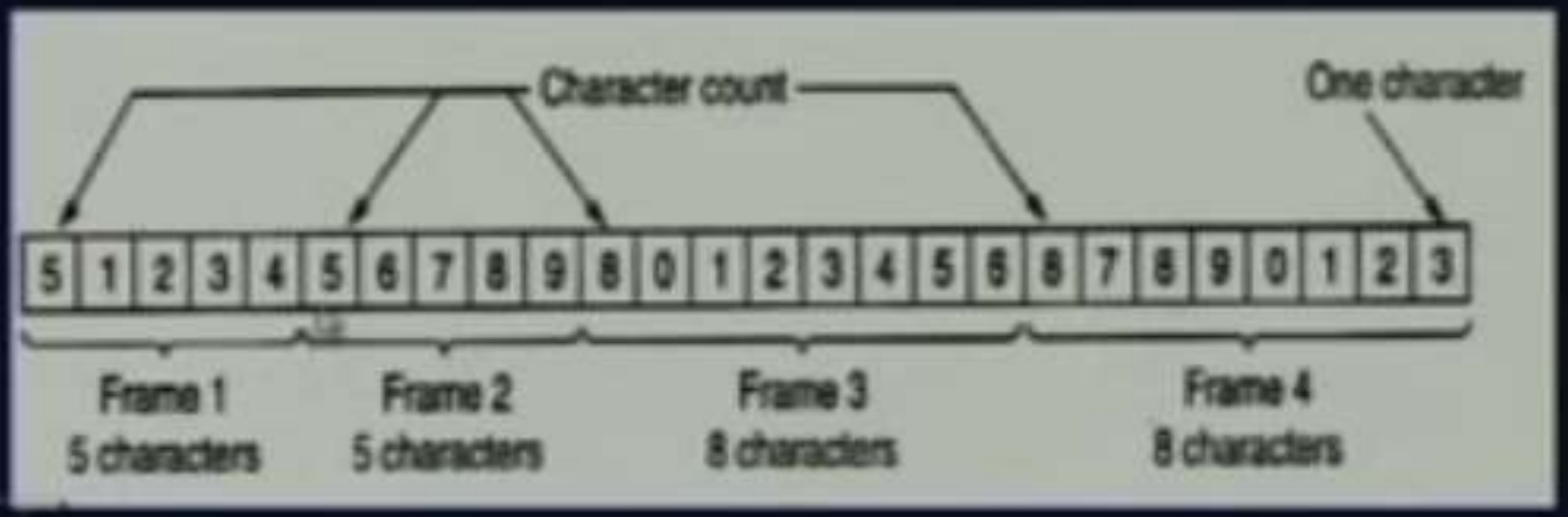
1. Length Count
2. Bit Stuffing
3. Character stuffing
4. Pulse encoding

Frames could be:

1. Fixed length (ATM)
2. Variable length
   - 1. Byte count (Decnet)
   - 2. Byte stuffing (SDLC)
   - 3. Bit stuffing (HDLC)
   - 4. Generic Framing Procedure
   - 5. Manchester Encoding(802.5)

Framing - Length Count

Make the first field in the frame's header be the length of the frame. That way the receiver knows how big the current frame is and can determine where the next frame ends.



Framing - Length Count

Disadvantage

- Receiver loses synchronization when bits become garbled. If the bits in the count become corrupted during transmission, the receiver will think that the frame contains fewer (or more) bits than it actually does.

Framing - Length Count

- Although checksum will detect the incorrect frames, the receiver will have difficulty resynchronizing to the start of a new frame.
- This technique is not used anymore, since better techniques are available.

Framing - Bit Stuffing

- Use reserved bit patterns to indicate the start and end of a frame.
- For instance, use the 4-bit sequence of 0111 to delimit consecutive frames. A frame consists of everything between two delimiters

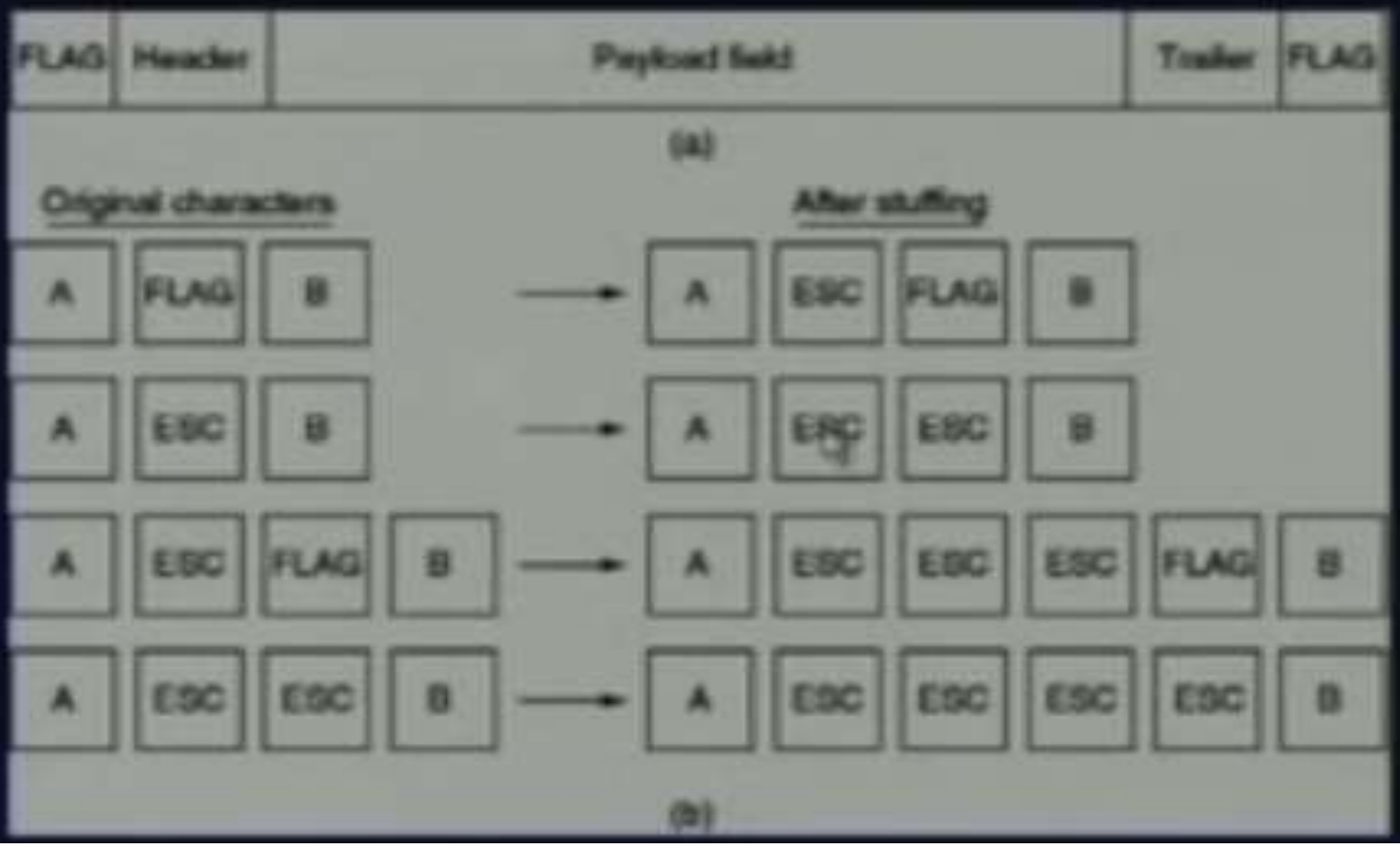
0111	frame	0111

Framing - Bit Stuffing

- Problem: What happens if the reserved delimiter happens to appear in the frame itself? If we don't remove it from the data, the receiver will think that the incoming frame is actually two smaller frames!

- Solution: Use bit stuffing. Within the frame, after every occurrence of two consecutive 1's insert 0. E.g., append a zero bit after each pair of 1's in the data. This prevents 3 consecutive 1's from ever appearing in the frame

Framing – byte stuffing



PPP – Point to Point Protocol

Bytes	1	1	1	1 or 2	Variable	2 or 4	1
	Flag 01111110	Address 11111111	Control 00000011	Protocol	Payload	Checksum	Flag 01111110

Point to Point Data Link Control

- one sender, one receiver, one link: easier than broadcast link:
  - no Media Access Control
  - no need for explicit MAC addressing
  - e.g., dialup link, ISDN line
- popular point-to-point DLC protocols:
  - PPP (point-to-point protocol)
  - HDLC: High level data link control (Data link used to be considered "high layer" in protocol stack!

PPP Design Requirements [RFC 1557]

- packet framing: encapsulation of network-layer datagram in data link frame
  - carry network layer data of any network layer protocol (not just IP) at same time
  - ability to demultiplex upwards
- bit transparency: must carry any bit pattern in the data field

PPP Design Requirements

- error detection (no correction)
- connection liveness: detect, signal link failure to network layer
- network layer address negotiation: endpoint can learn/configure each other's network address

PPP non-requirements

- no error correction/recovery
- no flow control
- out of order delivery OK
- no need to support multipoint links (e.g., polling)

Error recovery, flow control, data re-ordering all relegated to higher layers!

PPP Data Frame

- Flag: delimiter (framing)
- Address: does nothing (only one option)
- Control: does nothing; in the future possible multiple control fields
- Protocol: upper layer protocol to which frame delivered (eg, PPP-LCP, IP, IPCP, etc)

1	1	1	1 or 2	variable length	2 or 4	1
01111110	11111111	00000011	protocol	info	check	01111110
flag	address	control				flag

PPP Data Frame

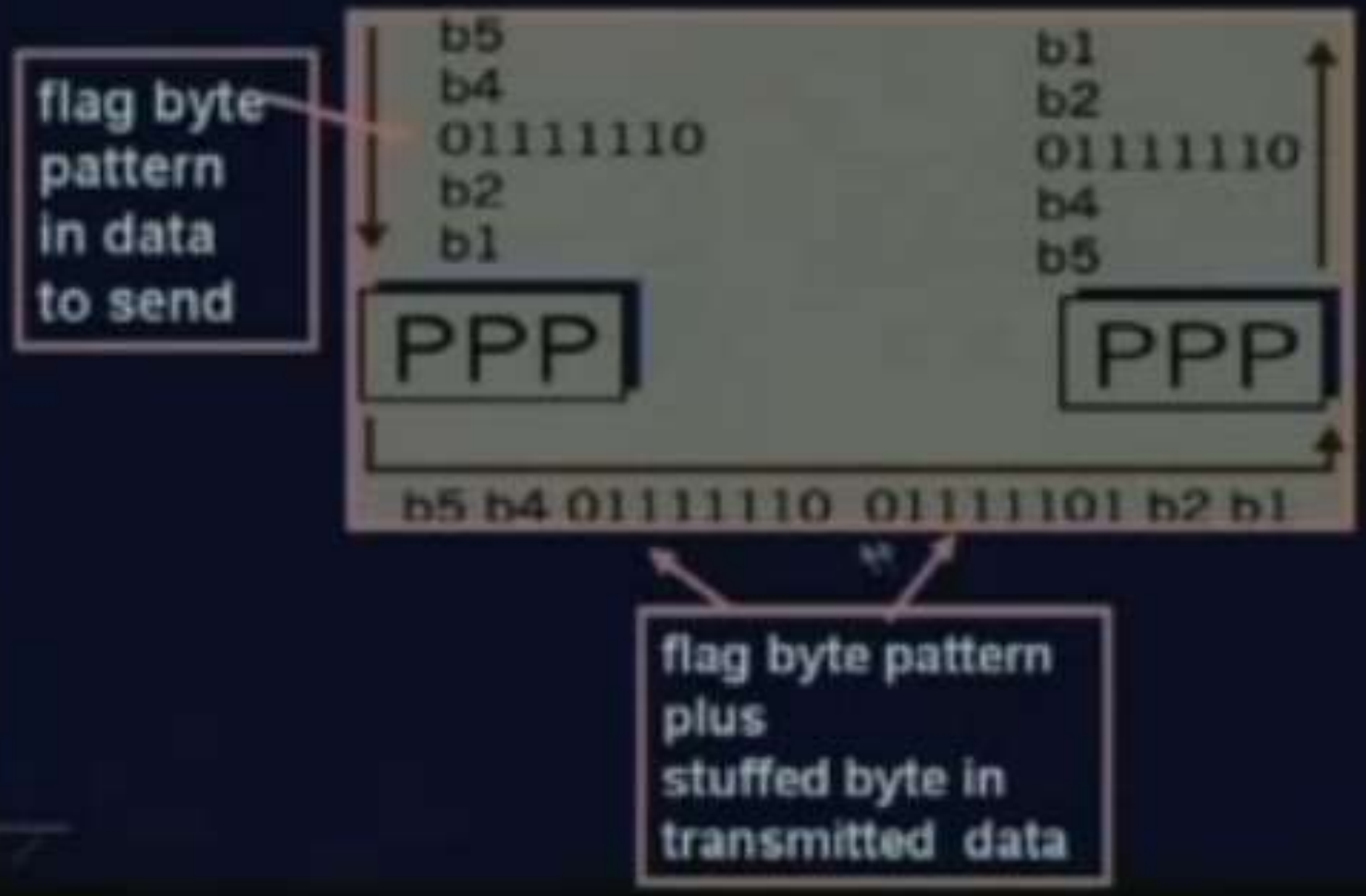
- info: upper layer data being carried
- check: cyclic redundancy check for error detection

1	1	1	1 or 2	variable length	2 or 4	1
01111110	11111111	00000011	protocol	info	check	01111110
flag	address	control				flag

Byte Stuffing

- "data transparency" requirement: data field must be allowed to include flag pattern <01111110>
  - Q: is received <01111110> data or flag?
- Sender: adds ("stuffs") extra < 01111110> byte after each < 01111110> *data* byte
- Receiver:
  - two 01111110 bytes in a row: discard first byte, continue data reception
  - single 01111110: flag byte

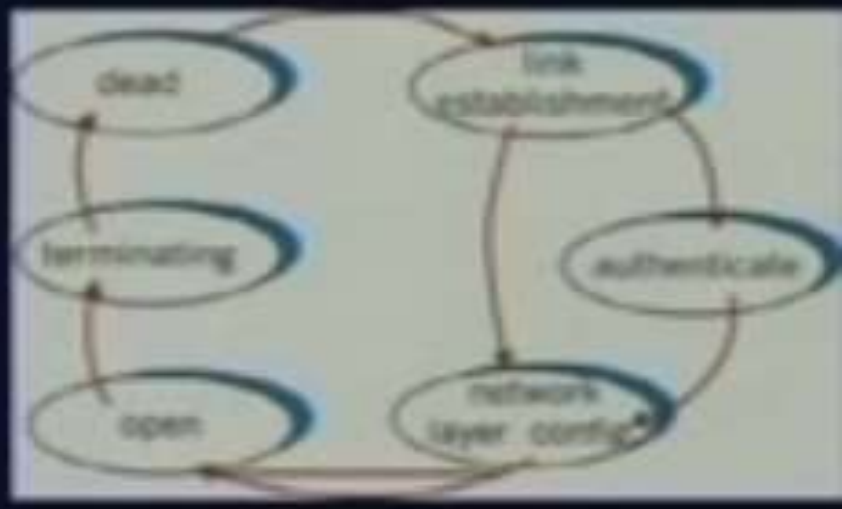
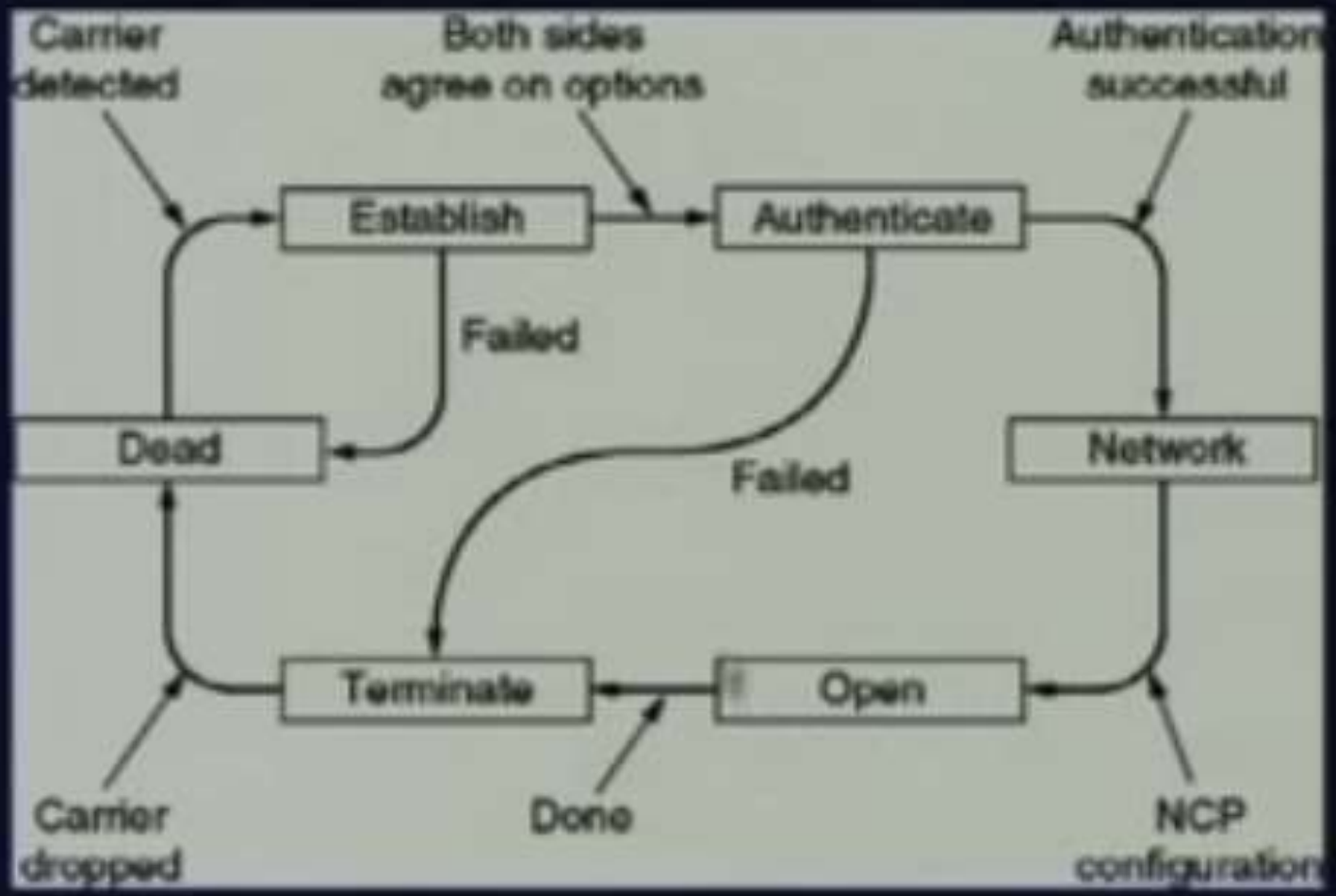
Byte Stuffing



PPP Data Control Protocol

Before exchanging network-layer data, data link peers must

- configure PPP link (max. frame length, authentication)
- learn/configure network layer information
  - for IP: carry IP Control Protocol (IPCP) msgs (protocol field: 8021) to configure/learn IP address



PPP – Point to Point Protocol